Bi-Hamiltonian Sturcture of Super KP Hierarchy

Source-backed arXiv sample

Abstract

We obtain the bi-Hamiltonian structure of the super KP hierarchy based on the even super KP operator $\Lambda = \theta^{2} + \sum^{\infty}_{i=-2}U_{i} \theta^{-i-1}$, as a supersymmetric extension of the ordinary KP bi-Hamiltonian structure. It is expected to give rise to a universal super W-algebra incorporating all known extended superconformal W_{N} algebras by reduction. We also construct the super BKP hierarchy by imposing a set of anti-self-dual constraints on the super KP hierarchy.

1 Source Metadata

This sample is based on arXiv record `hep-th/9109009`. Authors: Feng Yu.

2 Extracted Prose 1

We obtain the bi-Hamiltonian structure of the super KP hierarchy based on the even super KP operator math expression , as a supersymmetric extension of the ordinary KP bi-Hamiltonian structure. It is expected to give rise to a universal super math expression -algebra incorporating all known extended superconformal math expression algebras by reduction. We also construct the super BKP hierarchy by imposing a set of anti-self-dual constraints on the super KP hierarchy.

3 Extracted Prose 2

The integrable nonlinear (partial) differential KdV [1] and KP [2] equations are known to possess bi-Hamiltonian structures. This was first conjectured by Adler [3] and then proved by Gelfand and Dickey [4] for the KdV hierarchy, via the famous Gelfand-Dickey bracket. For the KP hierarchy, which is a set of multi-time evolution equations in Lax form